Insider Trading in the US Congress

2739941, 2852988, 2869153, 2841038,2766610, 2791195 and 2865081

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Set-up your environment

Title Page

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Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

Public confidence in members of congress' ability to act in the best interests of the country, uninfluenced by personal financial incentives, is essential to maintaining the integrity of democratic institutions. However, recent studies have brought up worries about possible conflicts of interest resulting from American congress members' stock market dealings. A significant study published by Abdurakhmonov et al. discovered that U.S. senators' stock purchases frequently resulted in abnormal positive returns, especially when the senators had jurisdiction over the companies through committee assignments or when the companies had made political connections, such as through lobbying or political donations (Abdurakhmonov et al., 2022). The Stop Trading on Congressional Knowledge (STOCK) Act of 2012 sought to address insider trading by forcing members of Congress to make prompt financial transaction disclosures, but enforcement has been difficult. For example, in the beginning of the COVID-19 pandemic in 2020, a number of senators came under fire for stock market activity that occurred after they had received classified briefings. The current regulatory system may have flaws, as evidenced by the fact that no charges have been brought despite the start of the current framework (Campaign Legal Center, n.d.). In response, legislative attempts have continued to these persistent concerns. By expressly defining and prohibiting such behavior, the Insider Trading Prohibition Act (H.R. 2655), which the House approved in 2021, seeks to strengthen prohibitions against trading on material, nonpublic information. The goal of this programming project is to evaluate publicly accessible financial disclosure data in order to find patterns that might point to insider trading by members of congress. By developing tools that analyze congressional financial disclosures and highlight possible illegal transactions, this programming project seeks to help correct that disparity. This initiative aims to increase the transparency of these financial activities by employing data science and programming methods to link transactions to market happenings, legislative changes, and committee assignments.

Part 2 - Data Sourcing

2.1 Load in the data

```
congresstrades <- read.csv("data/Copyofcongress-trading-all.csv")
senators_full_data<-read.csv("data/Copyofcongress-trading-all.csv")
SP_500_annual_returns <- read_csv("data/sp-500-historical-annual-returns.csv" , skip = 15)
##skip metadata
SP_500_annual_returns <- as.data.frame(SP_500_annual_returns) ##set data to class = dataframe
sp_500_monthly_full_data <- read.csv("data/sp500_monthly.csv")
```

2.2 Provide a short summary of the dataset(s)

```
head(as_tibble( senators_full_data))
```

```
## # A tibble: 6 x 19
##   Ticker  TickerType Company Traded Transaction Trade_Size_USD Status Subholding
##   <chr>   <chr>      <chr>   <chr>  <chr>       <chr>          <chr>  <chr>
## 1 NGL     ST         NGL EN~ Monda~ Sale        $15,001 - $50~ NEW    " EQUITAB~
## 2 FCX     ST         FREEPO~ Thurs~ Sale        $1,001 - $15,~ NEW    " MORGAN ~
## 3 V       ST         VISA I~ Thurs~ Purchase    $1,001 - $15,~ NEW    " TRUST O~
## 4 APPLE ~ ST         APPLE ~ Thurs~ Purchase    $360.00        NEW    " IRA ONE"
## 5 MSFT    ST         MICROS~ Thurs~ Sale        $100,001 - $2~ NEW    ""
## 6 BA      ST         BOEING~ Wedne~ Sale        $1,001 - $15,~ NEW    " BILL'S ~
## # i 11 more variables: Description <chr>, Name <chr>, Filed <chr>, Party <chr>,
## #   District <chr>, Chamber <chr>, Comments <chr>, Quiver_Upload_Time <chr>,
## #   excess_return <dbl>, State <chr>, last_modified <chr>
```

```
head(as_tibble(SP_500_annual_returns))
```

```
## # A tibble: 6 x 2
##   date        value
##   <date>      <dbl>
## 1 1928-12-31  37.9
## 2 1929-12-31 -11.9
## 3 1930-12-31 -28.5
## 4 1931-12-31 -47.1
## 5 1932-12-31 -15.2
## 6 1933-12-30  46.6
```

```
head(as_tibble(sp_500_monthly_full_data))
```

```
## # A tibble: 6 x 10
##   Date       SP500 Dividend Earnings Consumer.Price.Index Long.Interest.Rate
##   <chr>      <dbl>    <dbl>    <dbl>                <dbl>              <dbl>
## 1 1871-01-01  4.44     0.26      0.4                 12.5               5.32
## 2 1871-02-01  4.5      0.26      0.4                 12.8               5.32
## 3 1871-03-01  4.61     0.26      0.4                 13.0               5.33
```

```
## 4 1871-04-01  4.74     0.26      0.4                 12.6                5.33
## 5 1871-05-01  4.86     0.26      0.4                 12.3                5.33
## 6 1871-06-01  4.82     0.26      0.4                 12.1                5.34
## # i 4 more variables: Real.Price <dbl>, Real.Dividend <dbl>,
## #   Real.Earnings <dbl>, PE10 <dbl>
```

The Standard and Poor 500 annual return is sourced from Macrotrends (2025) .The annual percentage change of the S&P 500 index back to 1927. Performance is calculated as the % change from the last trading day of each year from the last trading day of the previous year.

The Standard and Poor 500 monthly return was originally compiled by Shiller (2005) and later cleaned and put into a .csv available to download on DataHub. Monthly dividend and earnings data are computed from the S&P four-quarter totals for the quarter since 1926, with linear interpolation to monthly figures. Dividend and earnings data before 1926 are from Cowles and associates, interpolated from annual data. Stock price data are monthly averages of daily closing prices through January 2000, the last month available as this book goes to press. The CPI-U (Consumer Price Index-All Urban Consumers) published by the U.S. Bureau of Labor Statistics begins in 1913; for years before 1913 1 spliced to the CPI Warren and Pearson's price index, by multiplying it by the ratio of the indexes in January 1913. December 1999 and January 2000 values for the CPI-U are extrapolated.

The Congress trading data cones from Quiver Quantitative (2025). It was cleaned and put into a .csv format available for download on Kaggle by user Shabba Rank. The excess return for each recorded trade is calculated as the difference between the security's Price Change Since Trade and the SPY 's change since the trade.

These are things that are usually included in the metadata of the data set. For your project, you need to provide us with the information from your metadata that we need to understand your data set of choice.

Limitations of the data sets:

One major problem of the data sets used in this research is that its data entries are not standardized. For example, date fields are frequently saved as simple text rather than as organized date objects, which can lead to mistakes during data cleaning and make temporal analysis more difficult. The data sets also includes a lot of incomplete data lines, due to missing disclosures or incorrect filings. By distorting the representation of trading activity and possibly underestimating the timing or frequency of insider trades among Members of Congress, these inconsistencies may greatly affect the validity of the results. It is crucial to handle these data problems in order to guarantee the reliability of any insights gained from the analysis.

2.3 Describe the type of variables included

The S&P 500 annual data contains two variables: a date variable as a character object column and a value column, containing the numeric return computed ass stated above for the year in the corresponding row of the date column.

The S&P 500 monthly data contains 10 variables. This includes date values and for each month the corresponding nominal and real prices, dividends and earnings, as well as Consumer price Index , PE10 and long interest rate.

The congress trading data contains 19 variables. This includes temporal and financial dat for individual trades made by congress members. Relevant variables for our analysis are the trade date, the congress members' names, the trade size, the transaction type, the excess return, political party, and the transaction type. These variables are character objects or numerics. This information is compiled from disclosures by congress members required by The US government under The Stock Trading on Congressional Knowledge Act.

Part 3 - Quantifying

3.1 Data cleaning

The congress trading data contained several duplicate entries. This inflated some variables, therefore we only kept distinct rows. We used the date of the transaction as the time variable in our calculations and plots, therefore, we had to convert the column entries from character objects to date objects to facilitate its use in grouping and plotting. For each transaction there is a trade size entry in USD. This entry is not a single numeric value but a range of possible trade sizes printed as a string of characters. We extract the numeric mean of the range boundaries to get an approximation of the trade size. We are only interested in the realised returns of Congress members, so we only keep "Sale" transactions, ignoring purchases and exchanges. We also eliminated rows that have missing transaction size entries or missing excess return entries.

We converted S&P 500 historical Montlhy returns date column from character objects to date objects. The month of October was not displayed properly in the data, showing up as 01 instead of 10 which caused inaccuracies in the plot, therefore, we made sure it was properly entered. We changed the SP500 column's name to "price", because its entries where the nominal price of the security for that month.

We converted the S&P 500 historical annual returns date column to only years. For the sake of clarity we changed the column name of the "value" column that contains the values of the s&p 500 return for that year to "SP_500_return".

3.2 Generate necessary variables

To facilitate grouping we create a trade year column and a trade month column. New variables are needed to produce a heat map plotting the percent factor by which the weighted average net return of congress members does outperform or under-perform the return of the market for each US State.

In a new data frame, we first calculate the weighted average excess return for all congress members from 2012 to 2024 grouped by state, by weighting each transaction's excess return with its transaction size, summing the weighted returns and then dividing them by the sum of all of the transaction sizes.

We then calculate the average market return from 2012 to 2024 by calculating the mean of the S&P 500 return values for these years. To get the weighted average net return for congress members per state we sum the S&P average return to the weighted average excess return for each state.

Finally to calculate how many times over the congress members' net return rate outperformed or under-performed the market rate we calculate the difference between the congress member's net rate and the market rate then divide it by the absolute value of the market rate for each state. To get the result as a percentage we multiply this ratio by 100.

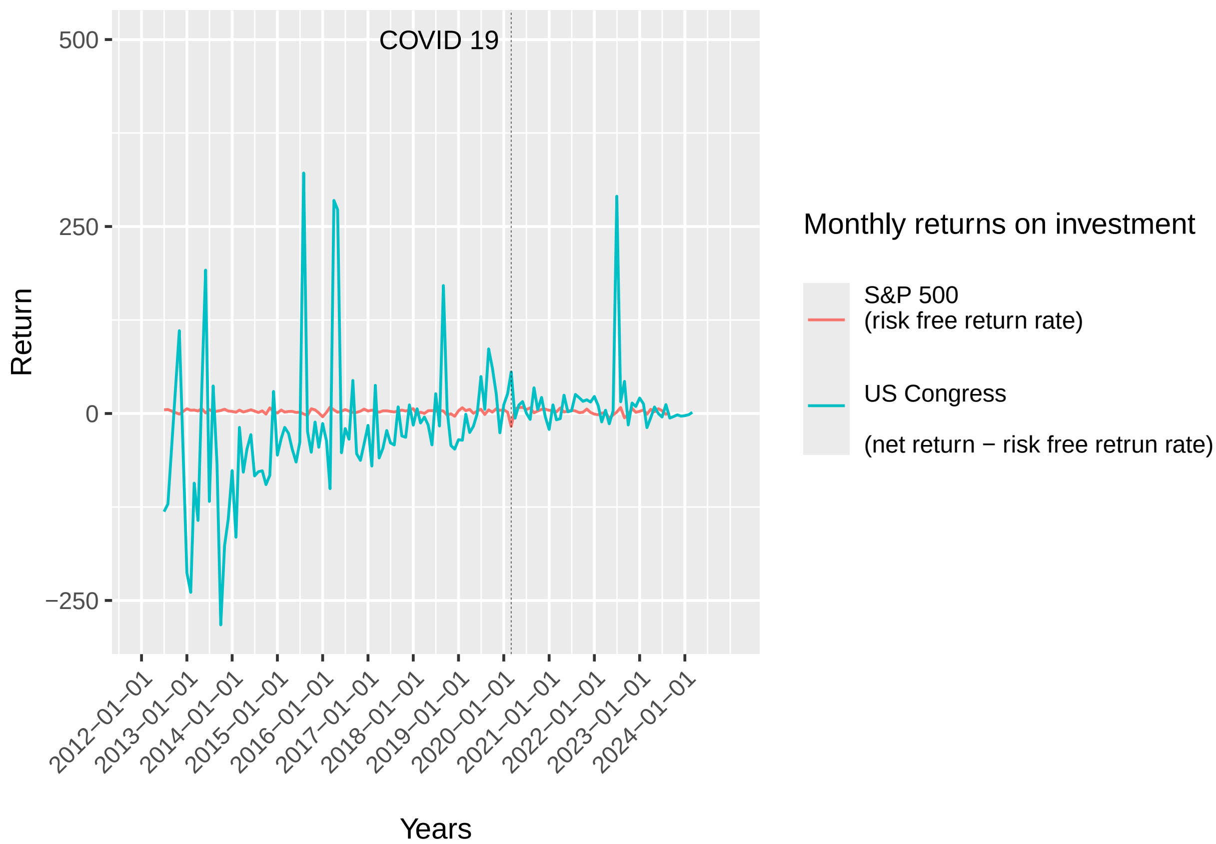
(If the congress rate is larger than the market rate we compute the difference as ***/net return - market return/***, otherwise we calculate the difference as ***-/market return - net return/***)

The weighted average excess return can also be calculated for each individual congress member. This can be used to separate the number of members who on average beat the market over the years *(weighted average excess return > 0)* from those who's trades on average returned less than the market *(weighted average excess return < 0)*. To differentiate between the two groups, we create a categorical variable that assigns "Higher than market" to members with positive excess returns and "Lower than market" to members with negative or zero excess return.

To get the market rate with a montlhy granularity for the temporal visualization we have to calculate the monthly market return rate. the return is equal to the price of the risk free security at the current period plus the dividend paid out minus the price of the security during the previous period all of which we divide by the price of the security during the previous period. We create this new variable in the S&P 500 monthly data set.

3.3 Visualize temporal variation

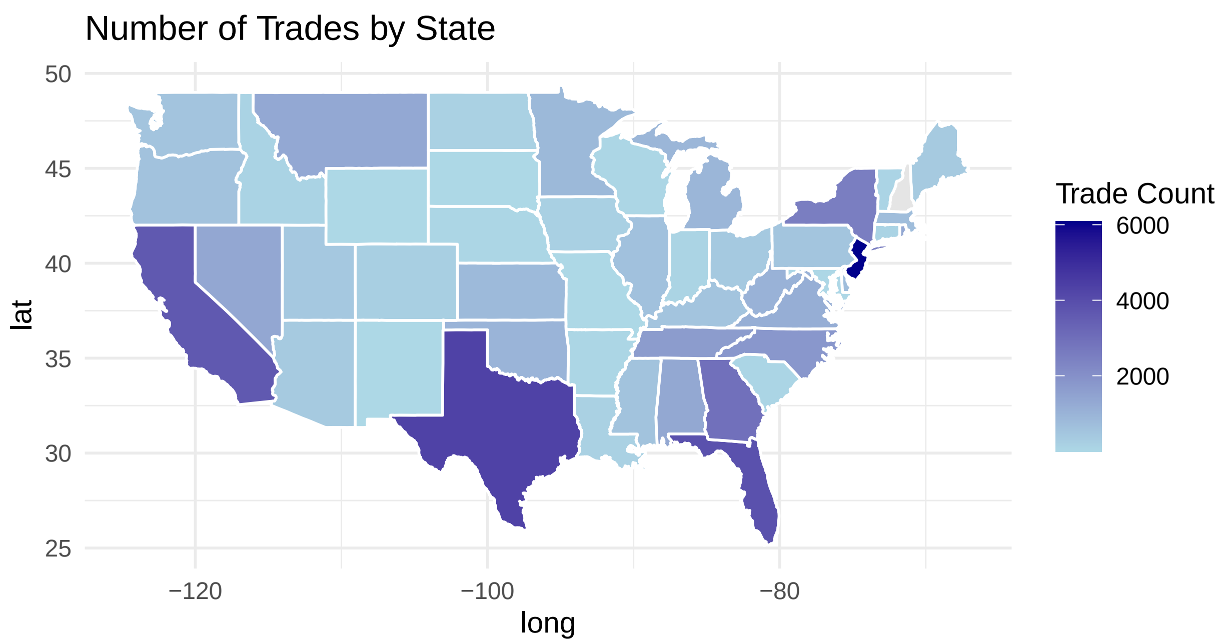
This plots the congress members' monthly weighted average excess return (net return - risk free return) and the market's risk free monthly return rate (S&P 500 return rate) from 2012 to 2024



This shows how the return of congress members' trades react to changes in the market through time. It can reveal if congress members can anticipate changes in the market, and show the difference between the excess return and the risk free rate.This is useful to get an idea of the net return of congress members at any point in time. This can reveal possible use of insider information.

3.4 Visualize spatial variation

```
ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill = num_trades)) +
  geom_polygon(color = "white") +
  coord_fixed(1.3) +
  scale_fill_gradient(low = "lightblue", high = "darkblue", na.value = "grey90") +
  labs(title = "Number of Trades by State", fill = "Trade Count") +
  theme_minimal()
```

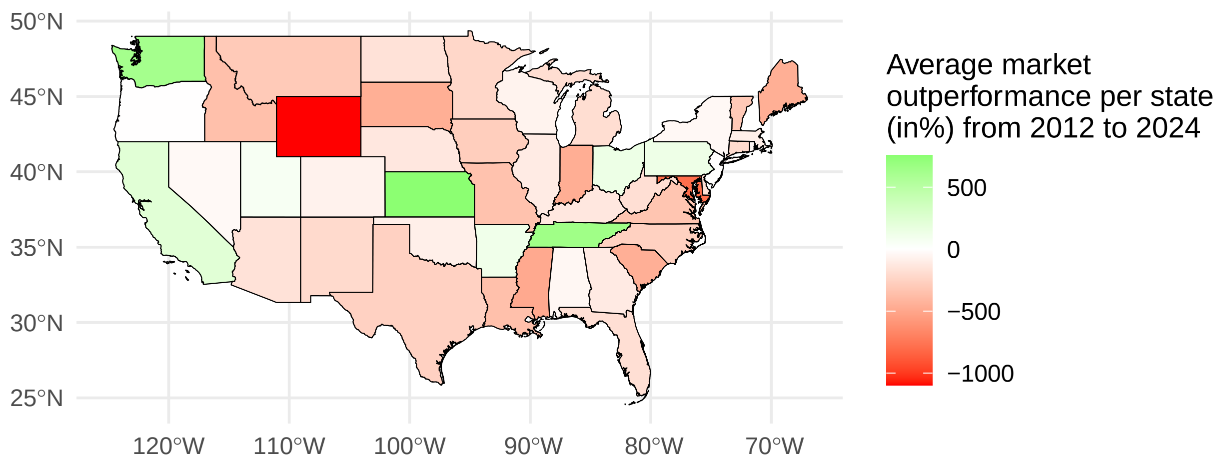


This graph is useful for a study on insider trading in the U.S. since it clearly displays the geographic distribution of congress trades by state, which may help spot unusual or trend-related behavior. For instance:

- Trade concentrations in certain states, such as New Jersey, California, Texas, and Florida, may indicate centers of political or economic activity where insider information is more easily available or regularly acted upon.
- Outliers, like New Jersey, which has an abnormally high trading activity but a relatively small number of representatives, may point to people or groups that should be looked at more closely. However, this will take a more comprehensive and complicated study.
- Comparative analysis across states can show whether insider trading is more common in areas with more representation, economic clout, or regulatory laxity.

```
#plot merged data
ggplot(us_states_returns_map, aes(fill = state_net_return_percentage)) +
  geom_sf( color = "black", size = 0.2 ) +

scale_fill_gradient2(low = "red", mid = "white", high = "green", midpoint = 0,
                     name = "Average market\noutperformance per state\n(in%) from 2012 to 2024") +
  theme_minimal()
```

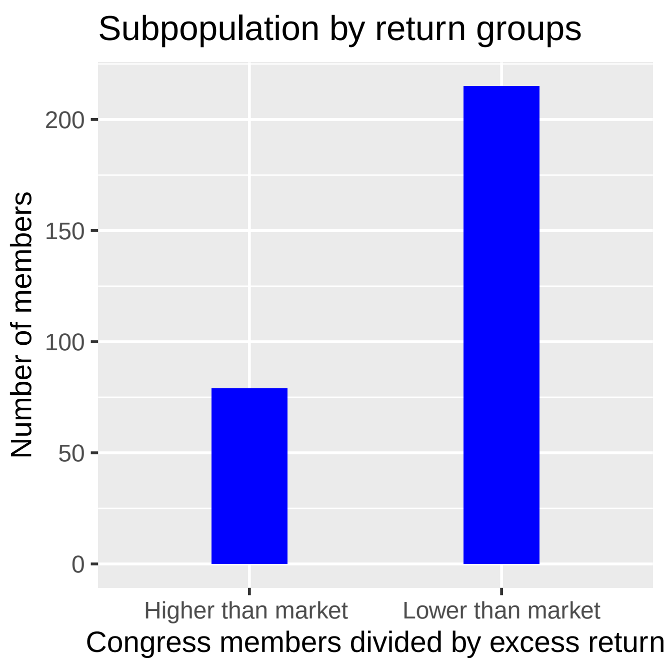


Grouping the congress members by state then calculating how many times over each state out-performed (under-performed) the market rate helps visualize if abnormal returns are spatially concentrated or if the trend is more widespread. A nationwide abnormal return pattern could indicate that congress members are capable of systematically achieving abnoraml returns which would point to insider trading.

3.5 Visualize sub-population variation

What is the number of congress members that beat the market from 2012 to 2024?

```
## Plot the number of congress members in each group
ggplot(senators_subpop_positive_excess_returns, aes(x = Group)
                                        ) +
  geom_bar(fill ="blue",
           width = 0.3,) +
  labs(x = "Congress members divided by excess return",
       y = "Number of members",
       title = "Subpopulation by return groups")
```



Plotting the number of congress members with higher return rates than the market and comparing it to the number of congress members with return rates lower than that of the market helps us understand whether the majority of is capable of achieving abnormal returns through insider trading. If there is a large proportion of congress members with positive excess returns that could hint at the presence of widespread access to insider information.

3.6 Event analysis

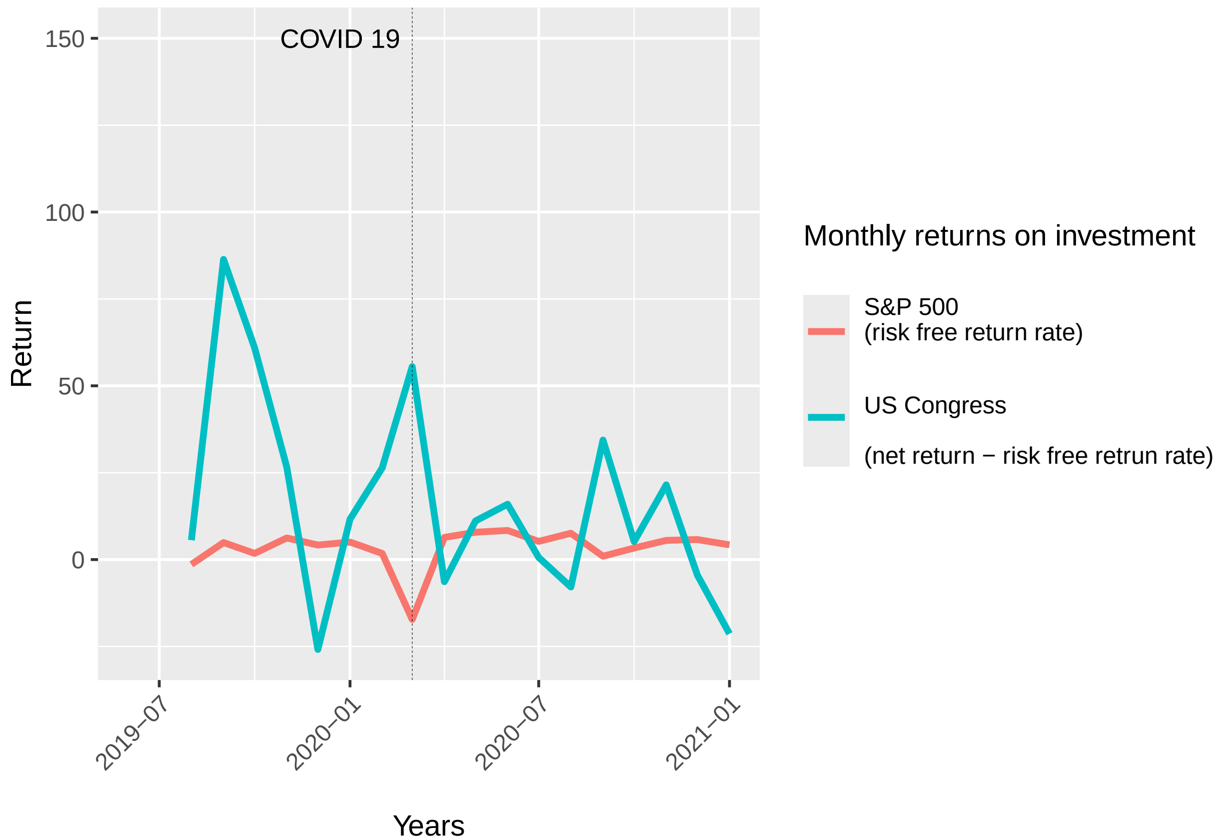
```
########################### covid 19 Zoom in line plot
reshaped_monthly_returns[165:206,] |>
  filter(!is.na(Return)) |> #remove NA returns
  ggplot( aes(x=Trade_month , y=Return, group=Series, color=Series)) +
  geom_line(linewidth = 1.2) +

  scale_x_date(
    name = "\nYears",
    limits = as.Date(c("2019-06-01", "2021-01-01")) ) +

  theme(
    axis.text.x = element_text(angle = 45, hjust = 1) ) +

  scale_color_discrete(name = "Monthly returns on investment\n",
                       labels = c("sp_500_monthly_return" = "S&P 500\n(risk free return rate)\n" ,
                                  "weighted_average_excess_return_monthly" =
                                    "\nUS Congress\n\n(net return - risk free retrun rate)")) +
  geom_vline(xintercept = as.Date("2020-03-01"),
             linetype = "dashed",
             color = "black",
             size = 0.1) +
```

```
annotate("text",  x = as.Date("2020-03-01"), y = 150, label = "COVID 19",hjust = 1.1,
         angle = 0,  size = 3.5, color = "black")
```



This plot focuses on a time interval surrounding the Covid 19 pandemic event. It is characterized by a local minimum in the market return rate. If congress members had prior knowledge of legislation and how it would affect the market leaing up to the Covid 19 pandemic event, we would observe an upward trend in the returns on investments made by congress members.

Part 4 - Discussion

4.1 Discuss your findings

Our investigation into potential insider trading among U.S. Congress members aimed to identify patterns of abnormal returns by comparing Senate trading activity with general market performance. When calculating the weighted average excess returns, we found that more than half of the Congress members generated negative weighted average excess returns, meaning their net returns were lower than those of the overall market. This finding challenges the notion that insider information is helping members achieve superior investment performance. A temporal analysis of congressional weighted average returns next to the risk free rate from 2012 to 2024 shows relatively random fluctuations in congressional returns, with notable spikes in 2015, 2016, and 2022. Excess returns failed to show any abnormal trend before and after the COVID 19 pandemic: weighted average excess return increased when the market return dipped and decreased when the market recovered, pointing to relatively stable average net returns throughout that period. When examining performance by state, Kansas, Tennessee, and Washington stood out with high average out-performance of the risk free rate. But the same cannot be said for the majority of states which performed below the market rate. Despite these insights, limitations such as incomplete or missing disclosures hindered the depth and precision of our analysis. To fully assess the extent and impact of potential insider trading in Congress, more comprehensive and transparent data is essential. Nevertheless, our findings highlight concerns about transparency and accountability in the financial activities of lawmakers.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/giovannivittorio/Programming_Project_Group4-4-

5.2 Reference list

Abdurakhmonov, M., Snider, R. E., Ridge, J. W., & Hasija, D. (2022). Perceptions of political Self-Dealing? An empirical investigation of market returns surrounding the disclosure of politician stock purchases. Strategic Management Journal. https://sms.onlinelibrary.wiley.com/doi/10.1002/smj.3459

Almeida, L. (2025, April 10). Trump ignites “insider trading” accusations after global tariffs U-turn. The Guardian; The Guardian. https://www.theguardian.com/us-news/2025/apr/10/donald-trump-ignites-insider-trading-accusations-after-global-tariffs-u-turn

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Himes, J. A. (2021, May 19). Text - H.R.2655 - 117th Congress (2021-2022): Insider Trading Prohibition Act. Www.congress.gov. https://www.congress.gov/bill/117th-congress/house-bill/2655/text

Data set References:

Macrotrends. (z.d.). S&P 500 Historical Annual Returns [Dataset]. https://www.macrotrends.net/2526/sp-500-historical-annual-returns

Shiller, R. (2005). Online Data: US Stock Markets 1871-Present and CAPE Ratio [Dataset] http://www.econ.yale.edu/~shiller/data.htm

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Quiver Quantitative (2025). Congress Trading [Dataset] https://www.quiverquant.com/congresstrading/

Shabba Rank. (2023). Congressional Trading (Inception to March 23) [Dataset]. Kaggle. https://www.kaggle.com/datasets/shabbarank/congressional-trading-inception-to-march-23?select=Copy+of+congress-trading-all+%283%29.csv